Presentation 2

Links to presentation(s) and code(s) on GitHub

- Link to All Calls Data Time Conversion Code
- Link to Presentation 2 Dashboard
- Link to Presentation 2

What did you do?

After receiving the objectives for Presentation 2, our team divided into two subgroups. Edwin and I focused on analyzing call redirections, specifically the fields Original reason, Related reason, and Redirect reason to better understand how calls flow through the system.

I focused on a **granular analysis of front desk calls** (where Called number = 13122296300) to identify how calls are being routed, answered, or redirected in the All Calls Dataset.

I began by revisiting the dataset to re-familiarize myself with the key columns relevant to redirection:

- Correlation ID (links multiple call legs together)
- Original reason, Related reason, Redirect reason (describe why redirection occurs)
- Redirecting number (shows who or what triggered the redirect)
- User and User type (indicate whether the call landed with a person, voicemail, queue, or menu)
- Answered (True/False flag, but can also show True if the call goes to voicemail)

I also helped design a visual walkthrough slide showing how a call can be broken down into three logical stages: (1) Call Start, (2) Call Movement, and (3) Call End. I then selected one real Correlation ID, '9108b891-9c8f-4056-a98c-dea3d0a121a8' (March 2025), and traced the call path through the dataset to illustrate redirection logic in practice. This confirmed how Webex "Follow Me" logic operates, when a call hits a user or number configured with Follow Me, the system automatically rings multiple destinations (e.g., other extensions, queues, or menus) until someone answers.

Next, I analyzed the front desk call performance through the following visuals. The original hypothesis was that many front desk calls might go unanswered. If this turned out to be true, then I could analyze what times of day the calls go unanswered. Since nearly all called were answered, I explored the time trend for where the calls were redirected most. To look at hour of day, I needed to create a new column to extract hour of day from the 'Start Time' column in the All-Calls Dataset. I also needed to convert the data from UTC to Chicago time. I did the following in python (see full code script linked above). I also added the new All Calls date for September 2025.

```
# Convert date to CST
df_main["Start time"] = df_main["Start
time"].dt.tz_convert("America/Chicago").dt.tz_localize(None)
# Extract hour of day
df_main["Start Time Hour"] = df_main["Start time"].dt.hour
print(df_main[["Start time", "Start Time Hour"]].head())
```

After that data cleaning was done, I created the following in my dashboard for this week.

1. **Answered vs Not Answered Front Desk (Donut Chart)**

- **X-axis:** Answer Indicator (True, False)
- **Y-axis:** Count of Correlation ID [Distinct]
- **Filter:** Called Number = 13122296300 (Front Desk), Direction = TERMINATING
- **Purpose:** Measures how many front-desk calls are ultimately answered versus unanswered.
- **Interpretation Note:** A "True" result in the Answer Indicator column can also include calls answered by automated systems (e.g., queues or voicemail), but for this subset, all final recipients were users, meaning calls were answered by a person.

2. **Redirect reason for Front Desk Calls (Bar Chart)**
    - **X-axis:** Redirecting reason
    - **Y-axis:** Count of Correlation ID [Distinct]
    - **Legend:** Answered (True/False)
    - **Filters:** Called Number = 13122296300, Direction = TERMINATING
    - **Purpose:** Identifies what the reasons for front desk redirection are
    - **Interpretation Note:** The dominant redirect reason is "Follow Me" meaning many of the calls from the front desk are automatically routed through Webex's sequential forwarding logic until answered.

3. **Redirecting Numbers from Front Desk (Bar Chart)**
    - **X-axis:** Redirecting Number
    - **Y-axis:** Count of Correlation ID [Distinct]
    - **Filters:** Called Number = 13122296300, Redirect Reason = Follow Me, Direction = TERMINATING
    - **Purpose:** Identifies where calls from the front desk are redirected most often.
    - **Interpretation Note:** This visual highlights that the majority of front desk calls (≈15k) are redirected to 13123411070 (Main Line), followed by smaller volumes to the Spanish (13125068647) and English (13125068646) menu lines.

4. **FollowMe Redirects by Hour of Day (Line Chart + Slicer)**
    - **X-axis:** Hour of Day (from Report Time, adjusted to CST)
    - **Y-axis:** Count of Correlation ID [Distinct]
    - **Filters:** Redirect Reason = FollowMe, Called Number = 13122296300 (Front Desk)
    - **Slicer:** Redirecting Number limited to top four destinations — 13123411070 (Main Line), 13125068647 (Spanish Menu), 13125068646 (English Menu), and 13123478342 (Other Line)
    - **Purpose:** Displays when front-desk calls are redirected throughout the day, segmented by redirect destination.
    - **Interpretation Note:** Redirects peak between 9 AM and 11 AM, particularly to the main line, reflecting the morning influx of client calls. Some redirections to translation menus (time trend varies).

How does it help the project?

While the data confirmed that the front desk was performing well (with over 97% of calls answered), I identified one area of improvement: Because a share of calls are redirected from the front desk number to the Spanish/English menu lines, it suggests a potential language routing gap. My recommendation was to add a bilingual routing option at the front desk, allowing English/Spanish selection earlier in the call flow to reduce unnecessary redirects and load on the main number. Additionally, my analysis showed that majority of front desk calls are redirected to the main line, which indicates that the Follow Me routing logic is functioning effectively and ensuring calls are received. However, this concentration of traffic also highlights consistent peaks between 9:00–11:00 AM, there could be an opportunity to consider additional staffing or call distribution strategies during these hours to maintain responsiveness. This is an area I plan to examine further to develop a more concrete operation recommendation.

Issues faced (if any) and attempts to resolve

One challenge I encountered was understanding the underlying Follow Me logic configured in Webex. For the front desk, I attempted to hypothesize how this sequence operates based on the data patterns. However, it is unclear whether the specific configuration of the Follow Me sequence can be accessed directly or if analysis and inference are the only ways to reconstruct it.

I also wanted to understand why some callers reach the front desk directly rather than the main line, since the front desk number does not appear to be publicly listed on Legal Aid's website. To explore this, I called the front desk myself and spoke with staff, who explained that callers often use this number to reach someone who can then transfer them to the registration line (which appears to correspond to the main line in the data). This helped clarify one possible use case for direct front-desk calls and provided context/confirmation for the redirection patterns observed. I would still like to ask Cynthia more about this.

Issues resolved (if any)

See above.

Next steps

Moving forward, one area I want to explore in more depth is the "Related Reason" field. While this analysis clarified that front-desk calls are generally being handled effectively, understanding the related reason codes could provide stronger insight into how calls are being transferred and why certain pathways occur (e.g., Deflective vs. Consultative Transfer). A useful next step would be to develop a drillable visualization in Power BI that traces the full pathway of front-desk calls, from the initial inbound call to the redirect destination, using correlation IDs as a connector. I could make a slicer to do this for not just front desk calls, but for all the top calls. I also want to combine my time analysis with Edwin's dashboard for Original and Redirect Reasons for all the top called numbers. I can build in hourly and weekly call trends for each major redirect path for these top numbers.

References (Mention if you built up on someone else's work)

- Column name/Description Documentation
- Muse Miao_Presentation1_Code (I used a line from this to help me convert time zones for the All Calls dataset)
- Call-number clarification dataset
- Professor Kirshna Import Code file for All Calls

Presentation 1

Links to presentation(s) and code(s) on GitHub

- Link to Presentation 1
- Link to Stat 390 Data and Power BI Visuals

What did you do?

Overall Note: While our team collaborated closely, reviewing each other's visuals, sharing feedback, and brainstorming insights together, this documentation focuses on the areas I led and explored most deeply, particularly the development and analysis of redirection-related metrics and visuals.

All Calls Data Cleaning: I processed the All-Calls data into a single CSV file using the import code provided by Prof. Krishna and emailed it to my team. I then loaded this into Power BI to start my analysis. When I opened the file in Power BI, I did the following to clean the data. I focused my data cleaning on the columns I knew I would be using for my visuals.

- I converted 'Called Number' to a text value.
- In the 'Redirect Reason' Column, I Applied Transform → Format → Trim to remove extra spaces and ensure consistent text formatting. I also cleaned up value names and improved readability (ie. changed "NoAnswer" to "No Answer").
- I filtered out all rows where Duration ≤ 0, since calls with zero or negative duration likely represent incomplete or failed call records.

Metric Development: I then started to develop out a metric for redirection efficiency. I came up with looking at **Redirect Success Rate (RSR)** to measure how effectively Legal Aid's phone system redirects incoming calls to the correct destination without unnecessary transfers or failures. I am defining a *redirected call* as referring to any call that, at some point during its lifecycle, was transferred, forwarded, or otherwise routed away from its original destination to another endpoint (such as a queue, another user, or voicemail).

The metric can be calculated as follows:

1. RSR – the percentage of redirected calls that end with a live person answering.
    a. RSR = # of redirected sessions with Success (Human)/number of redirected sessions (Correlation IDs) x 100

Before defining this metric, I spent time exploring the raw call data and reading through the Webex Calling detailed call-history documentation to understand how different columns interact. I specifically examined the relationships between 'Original reason', 'Redirect reason', 'Related reason', 'Answer indicator', and 'User type' across multiple rows. Through this process, I observed that:

- Calls can contain multiple legs under the same 'Correlation ID', meaning a single call can appear across several rows as it's redirected through the system.
- In some cases, 'Original Reason' and 'Redirect Reason' were blank, yet the Answer indicator showed "Yes-PostRedirection", signaling that the call did experience redirection even if it wasn't recorded in those fields.
- The 'User Type' column helped distinguish between calls answered by a person (e.g., User, CallCenterStandard) and those ending in voicemail or automated systems (e.g., VoiceMailRetrieval, AutomatedAttendantVideo).

I also looked at the call-number clarification dataset that maps specific phone numbers to their assigned lines or services. From this dataset, it is confirmed that calls ending with 13122296300 represent a human-answered endpoint (front desk). This could also be used in the classification logic for Success (Human) to supplement or replace User type fields with known human-answered numbers where system labeling may be incomplete or ambiguous. However, not all the numbers on this google sheet have descriptions so further work would need to be done on this documentation before using it.

I made the following assumptions based on all of this:

- Blank values for 'Original Reason' mean that no redirection occurred and blank values for 'Redirect Reason' mean no additional redirection occurred
    - This is true unless other evidence (e.g., "Yes-PostRedirection") indicates otherwise
- Each Correlation ID uniquely represents one full call session
- The final answered leg represents the true endpoint of the call
- The User type field reliably distinguishes human endpoints from automated or voicemail systems
- Some of the HuntGroup entries may represent real agents, but this needs confirmation before including them as "human" endpoints

Using these findings, I developed the logic for how to identify when a call qualifies as redirected and how to determine whether it ended successfully. A call session (identified by Correlation ID) is considered redirected if any leg of the call meets one or more of the following conditions:

- Original reason is not blank, or
- Redirect reason is not blank, or
- Related reason indicates redirection activity (e.g., Deflection, CallForward, CallQueue, HuntGroup, SequentialRing, SimultaneousRingPersonal, RoutePoint, Executive), or
- Redirecting number is not blank, or
- Answer indicator = "Yes-PostRedirection"

For success classification:

- Success (Human): The final answered leg (Answered = TRUE) has a User type ∈ {User, CallCenterStandard} (optionally including HuntGroup if this also represents an agent endpoint).

Once validated, I plan to visualize the RSR metric in Power BI and broken down by Redirect reason (e.g., *NoAnswer, CallQueue, UserBusy, Deflection, etc.*) to highlight where call routing performs well or fails. For example, low RSR for *NoAnswer* may indicate staffing or scheduling gaps or High RSR for *Deflection* suggests that automated attendants or routing rules are working effectively.

<u>Visuals:</u>

After developing the Redirect Success Rate (RSR) metric framework, I created several Power BI visuals to begin exploring the data and validating my assumptions. I also helped write the key insights for all of these listed in the official presentation.

- **Volume of Redirection Events by Redirect Type (Bar Chart)**
    - **X-axis:** Redirect Reason
    - **Y-axis:** Count of Redirect Reason
    - **Filter:** Direction = TERMINATING (includes only inbound, receiving legs)
    - **Purpose:** Displays how frequently each redirection type occurs in the dataset

        - **Interpretation Note:** Since each Correlation ID can have multiple call legs, this visual represents the *number of redirection events*, not unique calls
            - Filtering by Direction = TERMINATING ensures the analysis focuses on the *receiving side* of each redirection, where the call actually landed
- **Average Duration per Redirection Type (Bar Chart)**
    - **X-axis:** Redirect Reason
    - **Y-axis:** Average of Duration (seconds)
    - **Filters:** Direction = TERMINATING; Duration > 0
    - **Purpose:** Measures how long calls last once redirected to their final destination.
    - **Interpretation Note:** Duration is measured *per call leg*, not for the entire call session. This helps compare how long interactions last once they reach their redirected endpoints.
- **Recipients of All Calls (Donut Chart)**
    - **Legend:** User Type (e.g., WCCAdapter, User, VoiceMailRetrieval, VirtualLine, AutomatedAttendantVideo, Unknown)
    - **Values:** Distinct Count of Correlation ID
    - **Filter:** Direction = TERMINATING
    - **Purpose:** Identifies where all calls ultimately terminate, distinguishing between human-answered and automated destinations.
    - **Interpretation Note:** Each slice represents *unique call sessions* (distinct Correlation IDs) grouped by the type of recipient on the terminating leg. Using a distinct count of Correlation ID ensures that each call session is counted only once, even if it included multiple legs or redirections. Filtering to Direction = TERMINATING focuses the analysis on the *final receiving side* of the call, rather than intermediary routing steps.

How does it help the project?

This work establishes a clean, consistent dataset and a validated framework for measuring redirection efficiency within Legal Aid's phone system. As outlined more in the presentation, my analysis helps lead to specific recommendations about reviewing the call routing set up, strengthening intake availability and simplifying redirection paths. My analysis also helps lead to the insight that a high share of calls end in voicemail and the recommendation to optimize voicemail procedures as a focus area.

Issues faced (if any) and attempts to resolve

Issue	Attempt to resolve
Multiple legs can exist under one Correlation ID.	*After discovering this, I made a note in my dashboard that the visuals show redirection event volume/duration, not unique calls. Double counting still can reflect the overall routing load. For future work, I plan to aggregate at call-level for unique behavior.*
There were blank values in both the 'Original Reason' and 'Redirect Reason' columns.	*I assumed that blank values indicated the call was not redirected but would like to confirm this. I also reviewed related fields such as Answer Indicator and User Type to identify cases where calls may have been redirected even with blank reason fields. For instance, some calls showed "Yes–PostRedirection" in the*

	Answer Indicator field, suggesting redirection occurred despite missing reason data. I need clarification on why this is the case. My hypothesis is that this specific case happens when redirection happens automatically through system-level routing or user setting and isn't logged as a redirect reason.
The reference table mapping phone numbers to lines/services was incomplete.	*Not used officially in work this time; will need an updated version later to help identify human-connected calls.*
Only NoAnswer, FollowMe, Deflection, Unconditional,UserBusy, and Unavailable appeared as redirect reasons in the data. The official documentation lists others (CallQueue, HuntGroup, TimeOfDay, Explicitxxx, Implicitid, Unrecognized).	*Need client clarification on why these categories are missing.*
The documentation defines call not answered as "the party isn't available to take the call. CF/busy or Voicemail/busy." What does CF stand for here?	*Need client clarification on this.*

Issues resolved (if any)

See above.

Next steps

- Refine Data Logic and Validation
    - Create logic to aggregate multiple legs under a single Correlation ID to analyze unique calls
    - Verify whether blank Original Reason and Redirect Reason fields truly indicate no redirection or if automatic/system-level redirects are being excluded from logs.
- Enhance Classification
    - Once the phone-number mapping sheet is updated, incorporate it to strengthen classification of Success endpoints in RSR.
- Expand the Dashboard
    - Incorporate a Redirect Success Rate (RSR) visual to monitor redirection efficiency across different redirect types.
        - Introduce visuals like an average duration heatmap by redirect reason, redirect reason success stacked bar chart, etc.

References (Mention if you built up on someone else's work)

- Column name/Description Documentation
- Professor Kirshna Import Code file for All Calls